

Lecture 6 ERS312 Notes

Ocean Drilling Program

- Project Mohole: was to drill very deep in earth's mantle (was unsuccessful; it reached the crust)
- East and western part of Newfoundland was smashed together
- Opening and closing of ocean basins = Wilson Cycle

Deep Sea Drilling Project

- Drilling into ocean floor is technologically possible
- People more aware of plate tectonics during Project Mohole
- Brought to life to confirm theory of plate tectonics
- Glomar Challenger drilled 624 cores into ocean floor worldwide
- Showed that ice in Antarctica has been around for 38 million years
- Verified plate tectonics

ODP Scientific Highlights

- Helps us to better understand what drives climate and climate change
- Extinction event that wiped out the dinosaurs

Multiple Drilling Programs

- Shallow depth drilling and deeper

Arctic Coring Expedition

- North pole was sub-tropical at some point
- Around 45 million years ago was evidence that the North pole has been freezing up

Future of Ocean Drilling

- RV Mengxian goal was to drill the earth mantle

Ocean Floor Topography and Plate Tectonics

*Belief was that the bottom of the ocean is flat and featureless

Challenger Expedition

- To study ocean basins (took soundings of sea floor to see what depth of sea floor was)
- Man standing on side of boat and dropped in a weight attached to a rope and measured this
- First evidence for underwater mountains

Echosounders

- Send sound signal from a ship that bounces back on sea floor and recorded by a transducer
- Topography of ocean floor = Bathymetry

First Map of North Atlantic published in 1957

- Years of work
- Clearly shows that there is a very large under water mountain chain that has elongated features (central rift valley)
- 20 years later → 1st Map of World ocean floor bathymetry

Multibeam Bathymetry

- Emits array of depth sounds in different directions
- Important for scientific purposes, for fisheries, for safety
- Safe to have ships pass if shallow ocean region?
- NW passage was shallow ship route (avoided)
- Has been used to find flight headed to China that disappeared (was unsuccessful)

Atlantic Ocean-Passive Continental Margin

- Slowly down to basin until you get to mid-ocean ridge

- Plates are pushed apart and Atlantic ocean keeps growing (plates do not sink underneath continent)

Pacific Ocean-Active continental margin

- Flat ocean floor with seamounts then suddenly gets an ocean trench then very steep to the surface
- Plate disappears underneath the continental crust
- Active continental margin around Pacific Ocean while passive is mostly everywhere else

Pacific Ring of Fire

- Have earthquakes and volcanoes in this area (due to plate movement)
- Over time Pacific Ocean will get smaller and smaller

Continental Margin Features

- Continental shelf is in shallow waters
- Shelf break is the transition from continental shelf to continental slope
- Why is there a line pronounced at 120m? Glaciation
- During glaciation we lower our sea level (massive ice sheets on continents meaning water came from somewhere)
- Ocean water evaporates and moves over land, falls in areas that are cold enough as snow
- Snow/ice formed in winter will fall back in the ocean in summer
- Greenland and Antarctica has large ice sheets
- 20,000 years ago nobody living in GTA as it was ice filled

Pleistocene Sea Level curve

- Warm periods=interglacial (sea level like present day)
- Cold periods=glacial

Milankovitch Cycles

- Eccentricity: earth orbits around sun (almost perfect circle)
- Sometimes earth's orbit more orbit
- Obliquity (tilt) fluctuates between 22.5 and 24.5 degrees

- Orbital Precession (orientation of earth axis to different star constellations)

Modern Continental Shelf

- Shallow shelf regions are formed by materials that are transported by rivers and dumped onto shelves

Global Continental Shelf

- Lowering sea level by 120m
- Bering strait

Continental slope

- Sediment can be flushed down the continental slope down to the deep sea
- Continental slope is major conduit for sediment that is transported from shelf → deep sea

Monterey Bay Submarine Canyon

- Canyons are main way of getting sediment to deep sea
- Transport happens during events such as earthquakes (triggers sediments to rush down)

Turbidity Currents

- Characterized by very turbulent waters (high velocity currents)
- Denser than water because it has all the sand in it (this is how it is available to get to deep sea)
- Can erode into the shelf

Evidence found in Newfoundland

- Found out about turbidity currents
- Was a strong earthquake (destabilized sediments and triggered underwater landslide)
- Damage to cables
- **Steep** topography= things rush down quickly

- Continental rise is area that has shallow topography and is defined by sediment that is deposited by many turbidity currents

Abyssal plain

- Endless flat area
- Don't receive much sediment
- Sediment here is usually one that rains to sea surface that sinks to sea floor (consists of dust) *volcanic ash
- The drier a landscape is the more you can blow away material on the landscape
- Oceanic crust that is covered by sediment